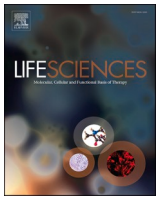
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Activation of mannose receptor C type 1 in macrophages improves renal fibrosis through mediating fibronectin endocytosis

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ABSTRACT

Aims: Excess extracellular matrix (ECM) deposition is the characteristic of renal fibrosis, owing to the imbalance between synthesis and degradation. Fibronectin could regulate the deposition of other ECM, thus plays a crucial role in the progression of renal fibrosis. Mannose receptor C type 1 (MRC1), largely expressed on macrophages, owns an extracellular fibronectin type II domain that binds to and internalizes collagen and thus involves in fibrosis modulation. The purpose of the present study was to investigate whether MRC1 participates in the internalization of fibronectin and whether alginate oligosaccharides (AOSC), a degradation product of alginate, has beneficial effects in the resolution of renal fibrosis via MRC1.

Materials and methods: Renal fibrosis models were constructed by unilateral ureteral obstruction (UUO) and unilateral ischemia-reperfusion injury (UIRI) in MRC1-WT and MRC1-KO mice. RAW264.7 cells were treated with TGF-β1 to induce pro-fibrotic responses. Expression of fibrotic markers and fibronectin endocytosis were examined.

Key findings: MRC1 gene knockout aggravated renal fibrosis in UUO and UIRI models. Inhibition of MRC1 exacerbated TGF-β1-induced pro-fibrotic responses in RAW264.7 cells. MRC1 regulated integrin β1-mediated fibronectin endocytosis through Arp2/3-Kindlin-2 signaling pathway. AOSC improved renal fibrosis by increasing MRC1 expression and endocytosis of fibronectin.

Significance: Our findings highlight the importance of MRC1 and fibronectin endocytosis in the development of renal fibrosis, suggesting that activation of MRC1 by AOSC is probably a therapeutic option to delay the progress of kidney fibrosis.

1. Introduction

Chronic kidney disease (CKD) is a clinical syndrome defined as irreversible structural damage and functional decline of the kidney, with a worldwide prevalence of 13–15 % [1]. Renal fibrosis is the most common outcome of CKD, characterized by excessive accumulation of extracellular matrix (ECM) due to imbalance between production and degradation [2]. Fibronectin is a major component of ECM and the first

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one to be deposited during fibrosis [3]. Fibronectin is an adhesive glycoprotein with various functional motifs by which fibronectin interacts with cell-surface receptors, collagens and fibrins, thereby modulating matrix organization and assembly, such as deposition and stability of other extracellular matrix proteins [4]. Fibronectin thus plays an important role in the development of kidney fibrosis.

Renal fibrogenesis is a dynamic process. One of the crucial defense pathways that minimizes renal fibrosis is degradation of extracellular matrix. Previous studies are focused on the extracellular pathway, in which ECM is degraded by matrix metalloproteinases via enzyme-catalyzed hydrolysis [5]. Intracellular processes also contribute to ECM turnover through receptor-mediated endocytosis [6]. This endocytic pathway is mainly carried out by two receptors called mannose receptor family [7] and integrins [8].

Mannose receptor C type 1 (MRC1) is the prototype member of lectin family. Extracellular region of MRC1 consists of an N-terminal CR (cysteine-rich) domain, a fibronectin type II repeat domain (FNII), and several C-type lectin-like domains (CTLDs) [9]. The FNII domain can bind and uptake collagen, internalized collagen is routed to lysosomes by endocytic vesicles for degradation [10]. MRC1 is mainly expressed in M2-like macrophages [11]. Macrophages are classified into classically activated macrophages (M1) and alternatively activated macrophages (M2). During the development of renal fibrosis, macrophages undergo a phase-dependent phenotypic change from M1 towards M2 that produces ECM components for tissue repair [12,13]. Hence, MRC1-expressing macrophages are one of the primary type of cells responsible for collagen internalization. Other members of MR family include MRC2, M-type PLA2R, and DEC-205 [14]. Given the reports that MRC2-expressing fibroblasts attenuate renal fibrosis via collagen turnover [15], we are interested in the function of MRC1 in macrophages and endocytosis of other matrix proteins during the progression of kidney fibrosis.

Integrins are another endocytic receptor that express in almost all cell types. These transmembrane cell adhesion receptors are formed by an α and β subunit [16]. There are at least three types of integrins which are laminin-binding integrins, collagen-binding integrins, and RGD-recognizing integrins [17]. They are involved in regulation of various cell functions, including cell proliferation, differentiation, survival, and matrix homeostasis through “outside-in” and “inside-out” signaling pathways [18,19]. Integrin β1 is a major receptor for fibronectin, which has been shown to promote the endocytosis and degradation of fibronectin in myofibroblast [20].

The therapeutic options of renal fibrosis are still limited, it is of great significance to seek drugs or therapeutic targets that delay the progression of renal fibrosis [21]. Alginate oligosaccharides (AOSC) are degradation products of alginate with the characteristics of low molecular weight, high stability, high water solubility and nontoxicity [22]. AOSC has a variety of biological activities, such as regulating immune response, anti-inflammation, antibacterial, anti-oxidation, anti-tumor and promoting cellular proliferation [23]. AOSC was shown to alleviate acute kidney injury induced by cisplatin by inhibiting oxidative stress [24]. In D-galactose-induced aging model, AOSC ameliorated kidney damage and fibrosis, although underlying mechanism remains unknown [25]. AOSC exerted a protective effect in busulfan-induced intestinal mucositis through MRC1 signaling pathway, suggesting that MRC1 likely be one of the targets of AOSC [26].

The purpose of the current study was to investigate a role and potential mechanism of MRC1 in preventing kidney fibrosis in unilateral ureter obstruction (UUO) and unilateral ischemia-reperfusion injury (UIRI)-induced mouse models. Our data showed that MRC1 attenuated kidney fibrosis likely by mediating fibronectin endocytosis through integrin β1. Additionally, we also found that AOSC alleviated renal fibrosis through interacting MRC1.

2. Methods

2.1. Animal models

Wild type (MRC1-WT) mice and MRC1 knockout (MRC1-KO) mice (C57BL/6 J background) were kindly provided by professor Dr. Jun Chen at Sun Yat-sen University. All mice were under a 12-h light/12-h dark cycle with water and chow ad libitum. All animal experiments were approved by Experimental Animal Ethics Committee at Sun Yat-sen University.

UUO Model: 10–12-week-old male mice were anesthetized with isoflurane. Mice were randomly divided into 4 groups: sham group of MRC1-WT mice, UUO group of MRC1-WT mice, sham group of MRC1-KO mice and UUO group of MRC1-KO mice. A median abdominal incision was performed to expose the left ureter, which was ligated with a 4–0 silk near the renal pelvis. The mice in sham groups were performed the same operation as those in UUO groups except no ligation on the left ureter [27]. Some mice were gavaged with 100 mg/kg AOSC (Haitang (Jiangsu) Biotechnology Co, Ltd., Nantong, Jiangsu, China) or saline every day after surgery. Mice were anesthetized with isoflurane at the 1st (1UUO), 3rd (3UUO) and 7th (7UUO) days after surgery and the left kidney tissues were harvested for further studies.

UIRI Model: 10–12-week-old male mice were anesthetized with isoflurane. Mice were randomly divided into 4 groups: sham group of MRC1-WT mice, UIRI group of MRC1-WT mice, sham group of MRC1-KO mice, UIRI group of MRC1-KO mice. A median abdominal incision was performed to expose the left kidney, and the left renal pedicles were dissected and clamped for 35 min using a microvascular clamp. At the 10th day after UIRI, the contralateral intact kidney was removed [28]. Some mice were gavaged with 100 mg/kg AOSC or saline every day after surgery. Mice were anesthetized with isoflurane at day 11 post-UIRI and the left kidney tissues were collected for further analyses.

2.2. Blood chemistry

At the end of the experiments, blood samples were collected from the vena cava for measurement of plasma creatinine levels. Creatinine concentrations of plasma samples were determined by using an enzyme immunoassay kit according to the manufacturer's instructions (BioAssay Systems, Hayward, CA).

2.3. Cell culture and treatment

Mouse leukemic monocyte/macrophage cell line (RAW264.7) was kindly provided by professor Dr. Jun Chen at Sun Yat-sen University. Cells were grown in DMEM containing 10 % FBS and 1 % penicillin/streptomycin. Human proximal tubular cells (HK2) were obtained from ATCC and grown in DMEM/F-12 containing 5 % FBS and 1 % penicillin/streptomycin. Normal rat kidney interstitial fibroblasts (NRK—49F) were kindly provided by professor Dr. Wei Chen at the First Affiliated Hospital of Sun Yat-sen university and grown in DMEM/F-12 containing 10 % FBS and 1 % penicillin/streptomycin. All cells were cultured at 37 °C in 5 % CO_2 atmosphere.

RAW264.7, HK2 and NRK-49F cells were seeded on 12-well plates and serum-starved for 12 h, respectively. The medium was changed to fresh serum-free DMEM or DMEM/F12 before treatment. Cells were pretreated with 1 and 2.5 mg/ml mannan (MRC1 inhibitor, sigma) for 1 h, and then treated with 1 or 2.5 ng/ml TGF-β1 for 24 h. MRC1 siRNA (RIBOBIO) was used to treat RAW264.7 cells for 24 h before stimulation with 1 ng/ml TGF-β1. RAW264.7 cells were incubated with 5 μM CK-666 (Arp2/3 inhibitor, MedChemExpress) or 1.25 μg/ml Anti-CD29 (integrin β1 inhibitory antibody, BD Pharmingen) for 1 h, and then treated with 1 ng/ml TGF-β1 for 24 h. RAW264.7 cells were pre-incubated with 1 mg/ml mannan, followed by incubated with 125 μg/ml AOSC, and treated with 1 ng/ml TGF-β1 for 24 h later. All the cells were collected for further protein or RNA analyses.

2.4. Western blot analysis

RIPA lysate and protease inhibitor cocktail were added to kidney tissues or cell samples, homogenized, left on ice for 10 min, and centrifuged at 12,000 rpm for 10 min at 4 °C. The supernatant protein concentration was assayed with the Pierce BCA Protein Assay (Thermo Fisher Scientific, Waltham, MA). Western blotting was performed by electrophoresis and incubation with primary antibodies (Table 1) overnight at 4 °C. The next day, membranes were incubated with horseradish peroxidase-labeled goat anti-rabbit IgG (H + L) (catalog 31,460, Thermo Fisher Scientific) and goat anti-mouse IgG (H + L) (catalog 31,430, Thermo Fisher Scientific) secondary antibodies for 1 h at room temperature. The blots were visualized in an ECL detection system (catalog FD8020, FDbio), and densitometric analysis was performed using AlphaEase software. The information of antibodies is presented in Table 1.

2.5. Histological analysis

Kidney tissues were fixed with 4 % buffered formalin followed by paraffin embedding, and then sliced into 4 μm sections. Paraffin-embedded kidney sections were stained with hematoxylin and eosin (H&E), Masson's trichrome staining and immunohistochemistry (IHC). Kidney sections used for IHC were dewaxed, rehydrated, incubated with 3 % H_2O_2 and then with the primary antibodies overnight at 4 °C (Fibronectin, 66,042–1, Proteintech, 1:300; Integrin β1, 12,594–1-AP, Proteintech, 1:300; F4/80, 70,076, Cell Signaling Technology, 1:300). The next day, sections were incubated with goat anti-rabbit IgG (H + L) (catalog 31,460, Thermo Fisher Scientific) and goat anti-mouse IgG (H + L) (catalog 31,430, Thermo Fisher Scientific) secondary antibodies for 1 h at room temperature, treated with diaminobenzidine, and counterstained with hematoxylin. Images were captured by a digital scanning microscope (Leica DM2000).

2.6. Immunofluorescence

RAW264.7 cells were cultured on coverslips (14 mm) and given different drug treatments. Cells were rinsed with PBS, fixed with 4 % paraformaldehyde, and then incubated with primary antibodies against MRC1 (ab64693, Abcam, 1:200) and integrin β1 (sc-9970, Santa Cruz, 1:50) overnight at 4 °C. Cells were incubated with secondary antibodies conjugated with Alexa Fluor 488 (goat anti-rabbit, catalog, Thermo Fisher Scientific, 1:400) and Alexa Fluor 555 (donkey anti-mouse, catalog, Thermo Fisher Scientific, 1:100) for 1 h at room temperature, nuclei were stained with DAPI (Thermo Fisher Scientific). Cells were imaged using a fluorescence microscope (Leica DMi8).

2.7. Fibronectin endocytosis assays

RAW264.7 cells were seeded on coverslips (14 mm) and given different treatments. The medium was changed to fresh serum-free DMEM before assay. Rhodamine Fibronectin (FNR01, cytoskeleton) was dissolved in ddH_2O, and added to medium (10 μg/ml). After a 1-h incubation under standard culture conditions, cells were rinsed three times with ice-cold PBS and fixed with 4 % formaldehyde solution for 15 min at room temperature. Cells were rinsed three times with TBST, and then stained with DAPI. Images were obtained using a fluorescence microscope (Leica DMi8) [29].

Table 1
Antibodies for western blot.

Target protein	Company	Cat. No	Dilution
Fibronectin	Boster	BA1772	1:2000
Collagen III	Abcam	ab184993	1:2000
TGF-β1	Santa Cruz	sc-130,348	1:1000
pSmad2/3	Cell Signaling Technology	8828	1:1000
Smad2/3	Cell Signaling Technology	8685	1:1000
MRC1	Abcam	ab64693	1:1000
Integrin β1	Proteintech	12,594–1-AP	1:1000
Integrin α5	Santa Cruz	sc-166,665	1:1000
Kindlin-2	Proteintech	11,453–1-AP	1:1000
Arp2	Proteintech	10,922–1-AP	1:1000
Arp3	Proteintech	13,822–1-AP	1:1000
GAPDH	Proteintech	60,004–1-Ig	1:10000

2.8. Quantitative real-time PCR

Total RNA from kidney tissues or RAW264.7 cells were extracted using TRIzol reagent (Invitrogen). The amount and quality of RNA were determined using a Nanodrop 2000 spectrophotometer (Thermo Scientific). 1 μg total RNA was reverse-transcribed to cDNA, and then quantified by real-time PCR using SYBR green master mix (AG-BIO) on the StepOnePlus system (Applied Biosystems). The sequences of the primers used are listed in Table 2.

2.9. Statistical analysis

Results are presented as the means ± SEM. Data were analyzed by One-way ANOVA and Student-Newman-Keuls tests for multiple comparisons. $P < 0.05$ was considered statistically significant. Values represent means ±SEM of three independent sets of experiments in in vitro studies. All data were analyzed and plotted using GraphPad Prism 6.0 software.

3. Results

3.1. MRC1 gene knockout aggravated kidney injury and fibrosis induced by 7UUO

The protein abundance of MRC1 in the obstructed kidneys of MRC1-WT mice were dramatically upregulated after 7UUO compared with sham, while MRC1 was slightly upregulated in the obstructed kidneys of

Table 2
Primer sequences for RT-qPCR (mouse).

Target gene	Primer sequence
MRC1 F	TCCTGGAGTCATAACTCTTTAGGGACA
MRC1 R	AAGCTGTTTCTTGTAGGGACCACTCA
Arg-1 F	CAGAAGAATGGAAGAGTCAG
Arg-1 R	CAGATATGCAGGGAGTCACC
iNOS F	TGGGTGAAAGCGGTGTTCTT
iNOS R	TAGCGCTTCCGACTTCCTTG
F4/80 F	CTTTGGCTATGGGCTTCCAGT
F4/80 R	GCAAGGAGGACAGAGTTTATCGTG
NGAL F	ACGGACTACAACCAGTTCGC
NGAL R	AATGCATTGGTCGGTGGGG
Integrin β1 F	ATGCCAAATCTTGCGGAGAAT
Integrin β1 R	TTTGCTGCGATTGGTGACATT
Arp2 F	GGAACTGAGGAGGAAGCG
Arp2 R	GGAACCCAAATGGAGAAT
Arp3 F	ACAGGAGGACGAGATGAT
Arp3 R	ACCACTTGCTGGGTTTAT
Kindlin-2 F	TCCATGTCACGGACCTGAAC
Kindlin-2 R	GCATCACCCCTCCGATGTG
AP-2 F	CCGACATCCGCAACTGTAAAA
AP-2 R	AGCTTCCATGTGTCCAAAGTC
Clathrin F	AGATTCTGCCCATTCGCTTTC
Clathrin R	TCAGTGCAATCACTTTGCTGG
Dynamin F	AGTCACCGGCACCAACAAG
Dynamin R	CTGGCAGGTCCACTAGAGTCA
Cortactin F	CTGTGTCCATCACGCAGGAT
Cortactin R	AGCGTCTGGTGTTCTTGGAAG
GAPDH F	TGACCTCAACTACATGGTCTACA
GAPDH R	CTTCCCATTCTCGGCCTTG

MRC1, mannose receptor C type 1; Arg-1, arginase-1; NGAL, neutrophil gelatinase-associated lipocalin; Arp2, actin-related protein 2; Arp3, actin-related protein 3; GAPDH, glyceraldehyde-3-phosphate dehydrogenase.

MRC1-KO mice than corresponding sham group (Fig. 1 A). The mRNA level of MRC1 was gradually increased in the kidney of MRC1-WT and MRC1-KO mice with 1UUO, 3UUO and 7UUO compared to their respective sham (Fig. 1B). These data likely indicated that MRC1 was probably involved in the development of UUO-induced fibrosis. The plasma level of creatinine was not changed among these groups (Fig. 1C). The mRNA level of a renal tubular damage marker NGAL were significantly upregulated in the obstructed kidneys from both MRC1-WT and MRC1-KO mice compared with corresponding sham mice, particularly mRNA level of NGAL in MRC1-KO mice was even higher than that in wild type mice (Fig. 1C). H&E staining showed that knockout of MRC1 was associated with exacerbated tubular and interstitial damage

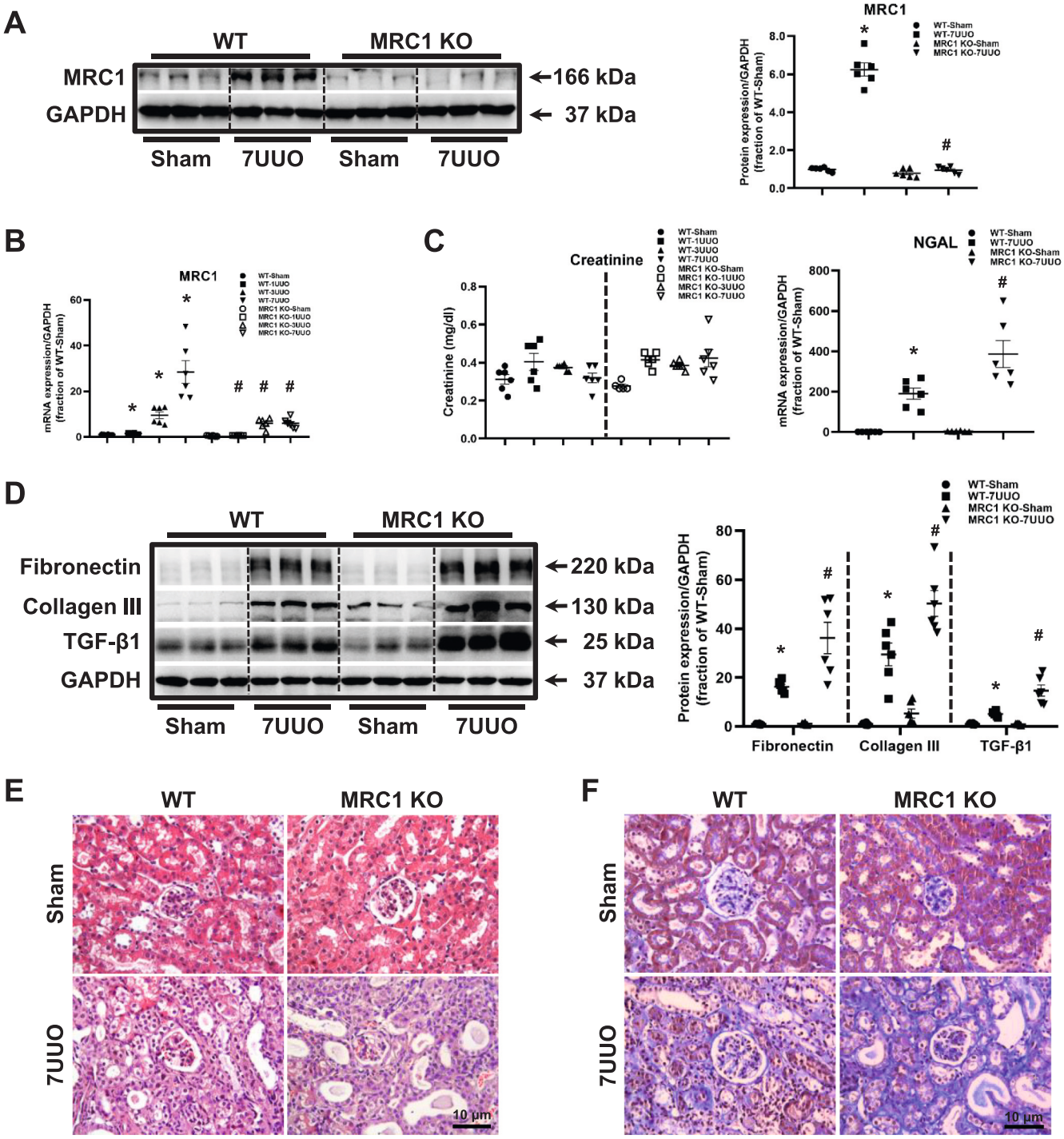


Fig. 1. MRC1 gene knockout aggravated kidney injury and fibrosis induced by UUO. (A) and (B) Representative immunoblots, corresponding densitometry analysis and mRNA level of MRC1 in the kidney of WT or MRC1-KO mice with or without 7UUO. (C) Plasma creatinine and mRNA level of NGAL in the kidney of WT or MRC1-KO mice with or without UUO. (D) Representative immunoblots and corresponding densitometry analyses of fibronectin, collagen III and TGF-β1 protein abundance in the kidney of WT or MRC1-KO mice with or without 7UUO. (E) and (F) Photomicrographs of hematoxylin-eosin staining and Masson's staining in the kidney of WT or MRC1-KO mice with or without 7UUO. Data are shown as mean ± SEM ($n = 6$ in each group), statistical analyses were performed using one-way ANOVA, *P < 0.05, when compared with WT-Sham group, $^{\#}$P < 0.05, when compared with WT-1/3/7UUO group. WT, wild type; KO, knockout; 1/3/7UUO, unilateral ureteral obstruction for 1/3/7 days.

in the obstructed kidney of MRC1-KO mice compared with MRC1-WT mice (Fig. 1E). The protein abundance of several fibrotic markers, including fibronectin, collagens III and TGF-β1 was markedly increased in the obstructed kidneys from MRC1-KO mice compared with WT mice (Fig. 1D). Masson's trichrome staining in the kidney confirmed severe fibrotic lesion after MRC1 knockout. Compared with sham group, 7UUO caused more collagen deposition in tubulointerstitial areas in the kidney, whereas the fibrotic lesion in the obstructed kidneys was more severe in MRC1-KO mice than that in MRC1-WT mice (Fig. 1F). Interestingly, the protein abundance of fibronectin and TGF-β1 was significantly increased in the obstructed kidneys from MRC1-KO mice compared with WT mice after 1UUO or 3UUO (supplementary data Fig. S1), indicating that the MRC1 is involved in regulation of fibronectin expression from the early stage of renal fibrosis. These results suggested that MRC1 deficiency aggravated kidney injury and fibrosis induced by 7UUO.

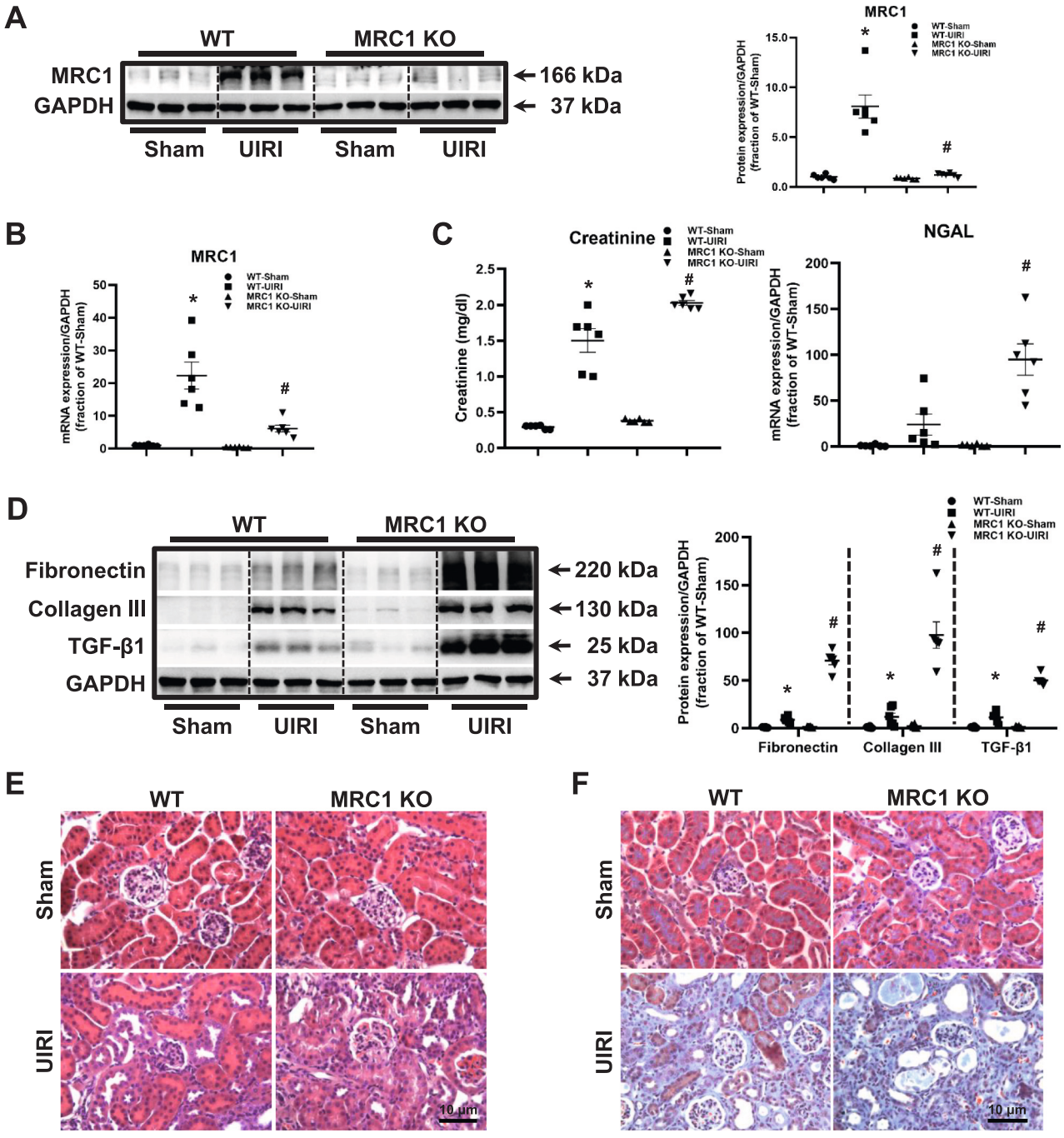


Fig. 2. MRC1 gene knockout aggravated kidney injury and fibrosis induced by UIRI. (A) and (B) Representative immunoblots, corresponding densitometry analysis and mRNA levels of MRC1 in the kidney of WT or MRC1-KO mice with or without UIRI. (C) Plasma creatinine and mRNA level of NGAL in the kidney of WT or MRC1-KO mice with or without UIRI. (D) Representative immunoblots and corresponding densitometry analyses of fibronectin, collagen III and TGF-β1 protein abundance in the kidney of WT or MRC1-KO mice with or without UIRI. (E) and (F) Photomicrographs of hematoxylin-eosin staining and Masson's staining in the kidney of WT or MRC1-KO mice with or without UIRI. Data are shown as mean ± SEM (n = 6 in each group), statistical analyses were performed using one-way ANOVA, *P < 0.05, when compared with WT-Sham group, #P < 0.05, when compared with WT-UIRI group. WT, wild type; KO, knockout; UIRI, unilateral ischemia reperfusion injury.

3.2. MRC1 gene knockout aggravated kidney injury and fibrosis induced by UIRI

To further examine the role of MRC1 in renal fibrosis, UIRI mouse model was also constructed in two strains of mice. Similarly, MRC1 protein and mRNA expression in the kidney was markedly upregulated after UIRI in MRC1-WT mice, but only moderately increased in MRC1-KO mice (Fig. 2A and B). Plasma creatinine level was markedly increased in both MRC1-WT and MRC1-KO mice after UIRI, even more in MRC1-KO mice when compared to MRC1-WT mice (Fig. 2C). The mRNA expression level of NGAL in UIRI models were higher in MRC1-KO mice than that in wild type ones (Fig. 2C). Consistently, protein expression of fibronectin, collagen III, and TGF-β1 in the kidney after UIRI was dramatically increased in MRC1-KO mice compared to MRC1-WT mice (Fig. 2D). Consistent with these, H&E staining (Fig. 2E) and Masson's trichrome staining (Fig. 2F) demonstrated more severe fibrotic lesion in the kidney of MRC1-KO mice than wild type mice. These data further supported a protective role of MRC1 in UIRI-induced renal injury and fibrosis.

3.3. Inhibition of MRC1 reduced endocytosis of fibronectin and aggravated TGF-β1-induced pro-fibrotic responses in RAW264.7 cells

All resident and infiltrated cells in the kidney, including tubular epithelial cells, macrophages and fibroblasts, participate in renal fibrogenesis after injury. As MRC1 was predominantly expressed in macrophages, we next examined whether MRC1 inhibition affected TGF-β1-induced pro-fibrotic responses in RAW264.7 cells. The protein abundance of MRC1, fibronectin and pSmad2/3 as well as the ratio of pSmad2/3 to Smad2/3 was markedly increased in response to TGF-β1 stimulation, while MRC1 inhibitor mannan (2.5 mg/ml) caused further increases in fibronectin and pSmad2/3 expression (Fig. 3A). The much-increased protein abundance of fibronectin was also observed when MRC1 was knockdown with siRNA in RAW264.7 cells stimulated by TGF-β1 (supplementary data Fig. S2).

MRC1 expressed on macrophages can bind and internalize collagen via its FN II domain, and deliver collagen to lysosomes where it is degraded [30]. To determine whether MRC1 mediates endocytosis of fibronectin, we added exogenous fluorescently labeled fibronectin into RAW264.7 cells. TGF-β1 increased the production of fibronectin (Fig. 3A), followed by an increase in endocytosis of fibronectin, which was seen in increased intracellular fluorescence intensity (Fig. 3B). When RAW264.7 cells were pretreated with mannan, a striking decrease in intracellular accumulation of fluorescently labeled fibronectin was found when compared with those treated with TGF-β1 alone (Fig. 3B). Together with increased fibronectin production, the decreased endocytosis of fibronectin caused by MRC1 inhibition likely contributed to a pro-fibrotic phenotype induced by TGF-β1 in RAW264.7 cells. To support this, we further examined the M2 polarization of macrophages after TGF-β1 treatment. TGF-β1 treatment was associated with increased the ratio of MRC1 to F4/80 and Arg-1 to F4/80, and decreased the ratio of iNOS to F4/80 (Fig. 3C), suggesting that TGF-β1 could induce the M2 polarization of macrophages. MRC1 inhibition by mannan significantly downregulated the ratio of MRC1 to F4/80 and Arg-1 to F4/80 compared to those treated with TGF-β1 alone, indicating a decreased proportion of M2 macrophages (Fig. 3C).

In addition, the protein expression of MRC1 was also detected in HK2 and NRK-49F cells treated with TGF-β1. There was no significant change in fibronectin protein expression after mannan treatment (Fig. 3D and E). Therefore, MRC1 expressed in M2 macrophages was probably play a significant role in TGF-β1-induced pro-fibrotic responses.

3.4. MRC1 gene knockout was associated with decreased expression of integrin β1 in fibrotic kidneys

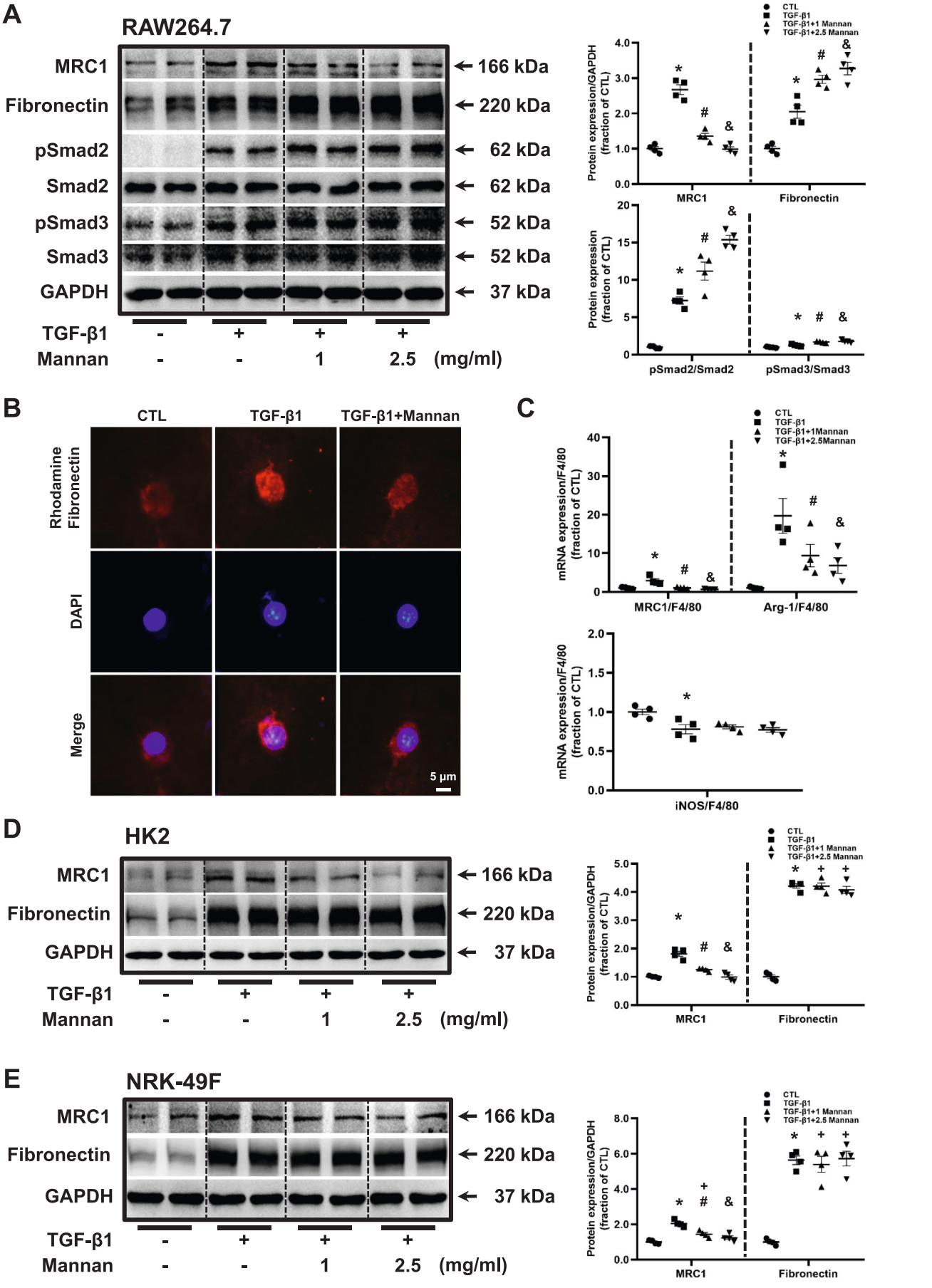
Next question would be by what mechanism MRC1 mediates fibronectin endocytosis? Integrin β1 is a surface adhesion receptor which is known to bind and internalize fibronectin, we presumed that MRC1 might modulate fibronectin endocytosis indirectly by integrin β1. Protein abundance of integrin β1 was dramatically increased in the kidney of both MRC1-WT and MRC1-KO mice after 7UUO (Fig. 4A) or UIRI (Fig. 4B), while MRC1 deficiency was associated with markedly attenuated upregulation of integrin β1 protein expression when compared with WT mice with 7UUO or UIRI (Fig. 4A and B). Integrins are heterodimers that consist of non-covalently associated α and β subunits, and α5 subunit is the most abundant one that interacts with integrin β1 to function as fibronectin receptors [31]. Similarly, MRC1-KO mice exhibited much less protein abundance of integrin α5 in the kidney after 7UUO or UIRI compared with MRC1-WT mice (Fig. 4A and B). Consistent with this, immunohistochemistry for integrin β1 showed less protein labeling in the kidney of MRC1 knockout mice subjected to 7UUO (Fig. 4C) or UIRI (Fig. 4D) compared to wild type mice. To better understand the presence and role of the MRC1 and integrin β1 pathway in macrophages in fibrotic kidney, immunohistochemical co-staining for integrin β1 and F4/80 was conducted on consecutive slices of MRC1-KO and MRC1-WT mice kidneys. Compared with sham groups, 7UUO caused much stronger staining intensity of F4/80 and integrin β1 on macrophages in the renal interstitium (Fig. 4E) of both two strains of mice, while the co-staining intensity was much weaker in the kidney of MRC1-KO mice than that in MRC1-WT mice (Fig. 4E). This data supported a potential interaction between MRC1 and integrin β1 in macrophages in kidney fibrosis.

3.5. MRC1 modulated expression of integrin β1 protein through Arp2/3-kindlin-2 signaling pathway

MRC1 could recruit a series of cytoskeleton-associated proteins and then lead to stabilization of actin-related protein 2/3 (Arp2/3) complex [32]. Arp2/3 complex can interact with kindlin-2 [33], which is supposed to activate integrin β1 by binding to the intracellular region of this fibronectin-binding receptor [34]. We therefore examined the Arp2/3-kindlin-2 signaling pathway in the fibrotic kidneys. Semiquantitative immunoblotting demonstrated remarkable increases in the protein expression of Arp2, Arp3 and kindlin-2 in the kidney of both wild type and MRC1 gene knockout mice subjected to 7UUO (Fig. 5A) or UIRI (Fig. 5B) compared with their corresponding shams. When compared to wild type mice with 7UUO or UIRI, the protein abundance of Arp2, Arp3 and kindlin-2 in the kidney was significantly decreased in MRC1-KO mice (Fig. 5A and B). mRNA levels of integrin β1, Arp2, Arp3 and kindlin-2 were remarkably upregulated in the kidney of MRC1-WT mice with 7UUO or UIRI compared with shams (Fig. 5C and D). mRNA levels of above proteins were also increased in MRC1-KO mice with 7UUO or UIRI, but not as much as those seen in MRC1-WT mice (Fig. 5C and D). The expression pattern of mRNA and protein of Arp2, Arp3 and kindlin-2 in the kidney was coincidentally with changes of integrin and fibrotic markers after MRC1 gene knockout, indicating that MRC1 is probably involved in ECM homeostasis during the development of fibrosis. Stabilization of Arp2/3 complex and its interaction with kindlin-2 by MRC1 activation depends on a series of upstream cytoskeleton-associated proteins (e.g. AP-2, clathrin, dynamin, and cortactin). mRNA levels of these molecules AP-2, clathrin, dynamin and cortactin were remarkably upregulated in the kidney of MRC1-WT mice with 7UUO or UIRI compared with shams (Fig. 5E and F), respectively, consistent with the alterations of mRNA expression of integrin β1, Arp2, Arp3 and kindlin-2.

3.6. MRC1 inhibition decreased expression of integrin β1 via Arp2/3-kindlin-2 signaling pathway in RAW264.7 cells

To examine whether MRC1 may interact to integrin β1, we identified their expression and distribution in RAW264.7 by immunofluorescence. There was a marked increase in expression and colocalization of MRC1 and integrin β1 in RAW264.7 treated with TGF-β1 (Fig. 6A),



(caption on next page)

Fig. 3. Inhibition of MRC1 decreased the endocytosis of fibronectin and aggravated TGF-β1-induced pro-fibrotic response in RAW264.7 cells. (A) Representative immunoblots and corresponding densitometry analyses of MRC1, fibronectin, pSmad2/3 and Smad2/3 protein abundance in RAW264.7 cells pretreated with mannan for 1 h followed by TGF-β1 treatment for 24 h. (B) Representative fluorescence microscopy images of RAW264.7 cells pretreated with mannan for 1 h followed by TGF-β1 treatment for 24 h and then incubated with Rh-FN. (C) mRNA levels of MRC1, iNOS and F4/80 in RAW264.7 cells pretreated with mannan for 1 h followed by TGF-β1 treatment for 24 h. (D) and (E) Representative immunoblots and corresponding densitometry analyses of MRC1 and fibronectin protein abundance in HK2 or NRK-49F cells pretreated with mannan for 1 h followed by TGF-β1 treatment for 24 h. Data are shown as mean ± SEM ($n = 4$ in each group), statistical analyses were performed using one-way ANOVA, $^{*}P < 0.05$ and $^{+}P < 0.05$, when compared with CTL group. $^{\#}P < 0.05$ and $^{\&}P < 0.05$, when compared with TGF-β1 group. CTL, control; TGF-β1, transforming growth factor-β1; Rh-FN, Rhodamine fibronectin.

interestingly, such a colocalization was much decreased in cells pretreated with MRC1 inhibitor mannan, indicating a potential interaction between MRC1 and integrin β1 (Fig. 6A). Moreover, TGF-β1 induced increases in abundance of integrin β1 and α5 protein in RAW264.7, which were moderately inhibited by mannan in a dose-dependent pattern (Fig. 6B) or by siRNA (supplementary data Fig. S2), indicating a modulation of integrin β1 by MRC1 in response to pro-fibrotic stimulation. Like seen in animals with MRC1-KO, inhibition of MRC1 by mannan caused significant decreases in protein abundance of Arp2, Arp3 and kindlin-2 in RAW264.7 treated with TGF-β1 (Fig. 6C). To clarify a role of Arp2/3-kindlin-2 signaling pathway in pro-fibrotic response, an Arp2/3 inhibitor CK-666 was used in RAW264.7 cells treated with TGF-β1. The protein abundance of Arp2, Arp3, kindlin-2 and integrin β1 was decreased after Arp2/3 inhibition by CK-666 (50 μM), while the protein expression of fibronectin was markedly increased (Fig. 6D), supporting our hypothesis. Taken together, these data suggested that MRC1 could regulate the expression and activation of integrin β1 in RAW264.7 cells likely through the Arp2/3-kindlin-2 signaling pathway.

3.7. Inhibition of integrin β1 reduced endocytosis of fibronectin and aggravated the pro-fibrotic response in RAW264.7 cells

To validate whether integrin β1 is involved in fibronectin endocytosis and pro-fibrotic responses, we used function blocking anti-integrin β1 antibodies (anti-CD29) to block integrin in RAW264.7 cells. Anti-CD29 significantly prevented upregulation in protein expression of integrin β1 and integrin α5 induced by TGF-β1, whereas protein expression of fibronectin and pSmad2/3 were significantly enhanced after anti-CD29 treatment compared with negative IgG control (Fig. 7A), which was similar to seen in the effect of MRC1 inhibitor mannan (Figs. 3A and 6B). Suppression of integrin β1 by anti-CD29 didn't affect the expression of MRC1 (Fig. 7A), in contrast, inhibition of MRC1 by mannan caused a significant decrease in protein abundance of integrin β1 (Fig. 6B), suggesting that MRC1 might be an upstream molecule of integrin β1 signaling in RAW264.7 cells. Fluorescence staining showed that anti-CD29 substantially reduced fibronectin endocytosis in RAW264.7 (Fig. 7B). These data suggested that integrin β1 appeared required for fibronectin endocytosis and pro-fibrotic response in macrophages.

3.8. AOSC upregulated MRC1 expression and attenuated renal fibrosis in mice with 7UUO or UIRI

Next, we aimed to examine whether AOSC protect kidney from fibrosis and potential mechanisms. H&E staining showed that the histopathological changes in renal histology were significantly reversed by AOSC treatment in MRC1-WT mice with 7UUO (Fig. 8A) or UIRI (Fig. 9A). Masson's staining and immunohistochemistry showed that AOSC reduced collagen deposition and weakened the staining of fibronectin in fibrotic kidneys (Fig. 8A and 9A, respectively). The protein abundance of fibronectin and collagen III were also obviously attenuated in the kidneys from 7UUO (Fig. 8B) or UIRI (Fig. 9B) mice treated with AOSC. Interestingly, in wild type mice with 7UUO or UIRI, protein expression of MRC1 and integrin β1 and α5 in the kidney were markedly upregulated in response to AOSC treatment (Figs. 8B and 9B, respectively). These results likely suggested a protective role of AOSC in preventing development of kidney fibrosis.

In contrast, AOSC failed to decrease the protein abundance of fibronectin and collagen III in the kidneys of MRC1-KO mice subjected to either 7UUO (Fig. 8C) or UIRI (Fig. 9C). MRC1 knockout also abrogated AOSC-induced up-regulation of renal integrin β1 and α5 in mice with 7UUO (Fig. 8C) or UIRI (Fig. 9C). These data supported the idea that AOSC may mitigate kidney fibrosis through MRC1 signaling pathway.

3.9. AOSC upregulated the expression of MRC1 protein and inhibited the pro-fibrotic responses in RAW264.7 cells

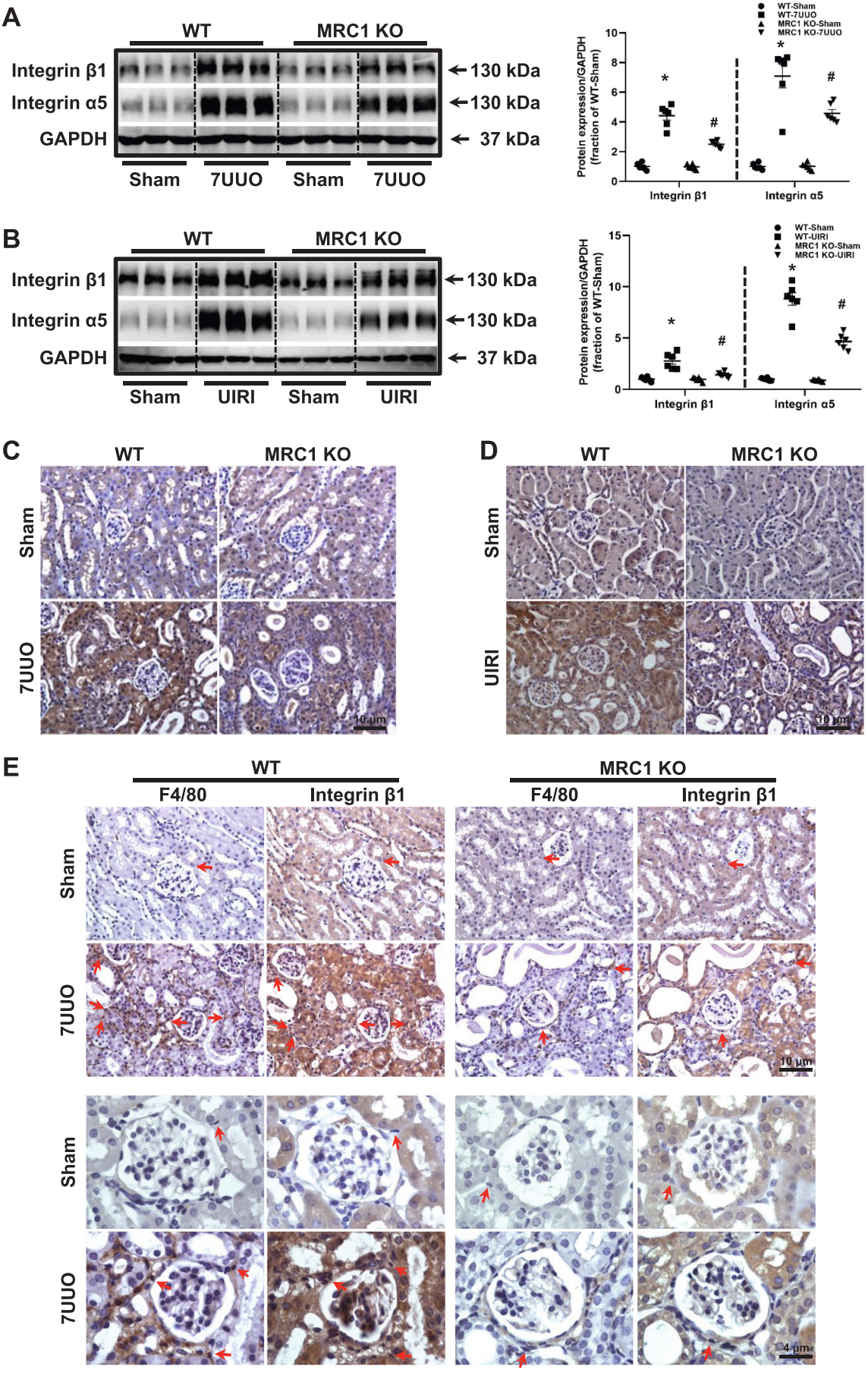
Under stimulation with TGF-β1, RAW264.7 cells pretreated with AOSC exhibited higher protein levels of MRC1 and integrin β1 than those with TGF-β1 treatment alone, whereas protein abundance of fibronectin was markedly decreased. While, when MRC1 was inhibited by mannan, AOSC pretreatment failed to significantly affect protein expression of MRC1 and integrin β1, and couldn't prevent the increase of fibronectin protein expression induced by TGF-β1 in RAW264.7 cells (Fig. 10A).

Finally, we determined whether AOSC ameliorated the pro-fibrotic responses by regulating fibronectin endocytosis in RAW264.7 cells. Fluorescence staining demonstrated that fibronectin endocytosis was dramatically increased in cells pretreated with AOSC compared with those treated with TGF-β1 alone, while this increased endocytosis of fibronectin was obviously disappeared after MRC1 inhibition by mannan (Fig. 10B). These results suggested that AOSC may suppress TGF-β1-induced pro-fibrotic responses in RAW264.7 likely by increasing MRC1-mediated fibronectin endocytosis.

4. Discussion

In the current study, we demonstrated that MRC1 in macrophages ameliorated renal interstitial fibrosis likely by interacting with integrin β1 which mediated fibronectin endocytosis. AOSC improves kidney fibrosis probably by interacting with MRC1. Several lines of evidence support this conclusion: i) Kidney fibrosis induced by UUO or UIRI was more severe in MRC1-KO mice than wild type mice. Inhibition of MRC1 in RAW264.7 cells exacerbated TGF-β1-induced pro-fibrotic responses. ii) MRC1 knockout or inhibition was associated with decreased integrin β1-mediated fibronectin endocytosis likely due to suppression of the Arp2/3-kindlin-2 signaling pathway. iii) AOSC improved renal fibrosis probably by up-regulating MRC1 expression and increasing integrin β1-mediated fibronectin endocytosis.

MRC1 is a transmembrane receptor that involved in immunity and maintaining homeostasis through receptor-mediated endocytosis and phagocytosis [35]. Extracellular region of MRC1 contains a N-terminal CR domain, a FNII domain and eight C-type lectin-like domains. The FNII domain can bind collagen [36] (Fig. 11). As a rapidly growing body of evidence shown, MRC1 performs a crucial role in the development of fibrosis through promoting collagen internalization [30]. In the present study, we found that MRC1 expression was minimal in normal kidneys, while in the kidney of mice with 7UUO or UIRI, the protein abundance of MRC1 was significantly upregulated, which was associated with marked fibrosis as seen increased protein expression of TGF-β1 and fibronectin. Interestingly, in MRC1-KO mice with 7UUO or UIRI, the protein expression levels of fibronectin and collagen III in the kidneys were much higher than that in WT mice. These findings provide



(caption on next page)

Fig. 4. MRC1 regulated expression of integrin β1 in 7UUO or UIRI models. (A) and (B) Representative immunoblots and corresponding densitometry analyses of integrin β1 and integrin α5 protein abundance in the kidney of WT or MRC1-KO mice with or without 7UUO (A) or UIRI (B). (C) and (D) Photomicrographs of immunohistochemistry of integrin β1 in the kidney of WT or MRC1-KO mice with or without 7UUO (C) or UIRI (D). (E) Photomicrographs of immunohistochemistry for F4/80 and integrin β1 in the kidney of WT or MRC1-KO mice with or without 7UUO. Data are shown as mean ± SEM (n = 6 in each group), statistical analyses were performed using one-way ANOVA, *P < 0.05, when compared with WT-Sham group. #P < 0.05, when compared with WT-7UUO or UIRI group, respectively. WT, wild type; KO, knockout; 7UUO, unilateral ureteral obstruction for 7 days; UIRI, unilateral ischemia reperfusion injury.

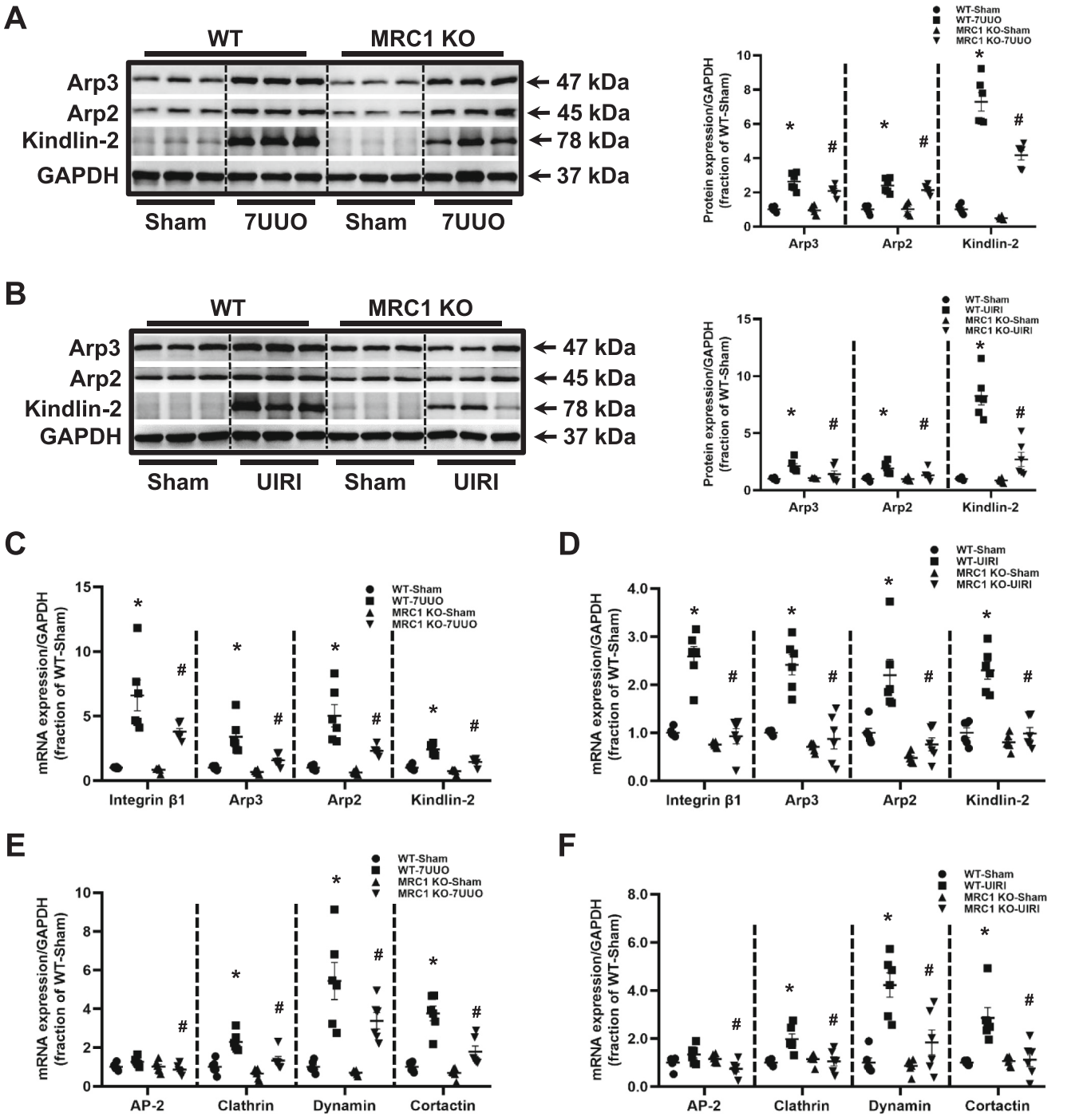
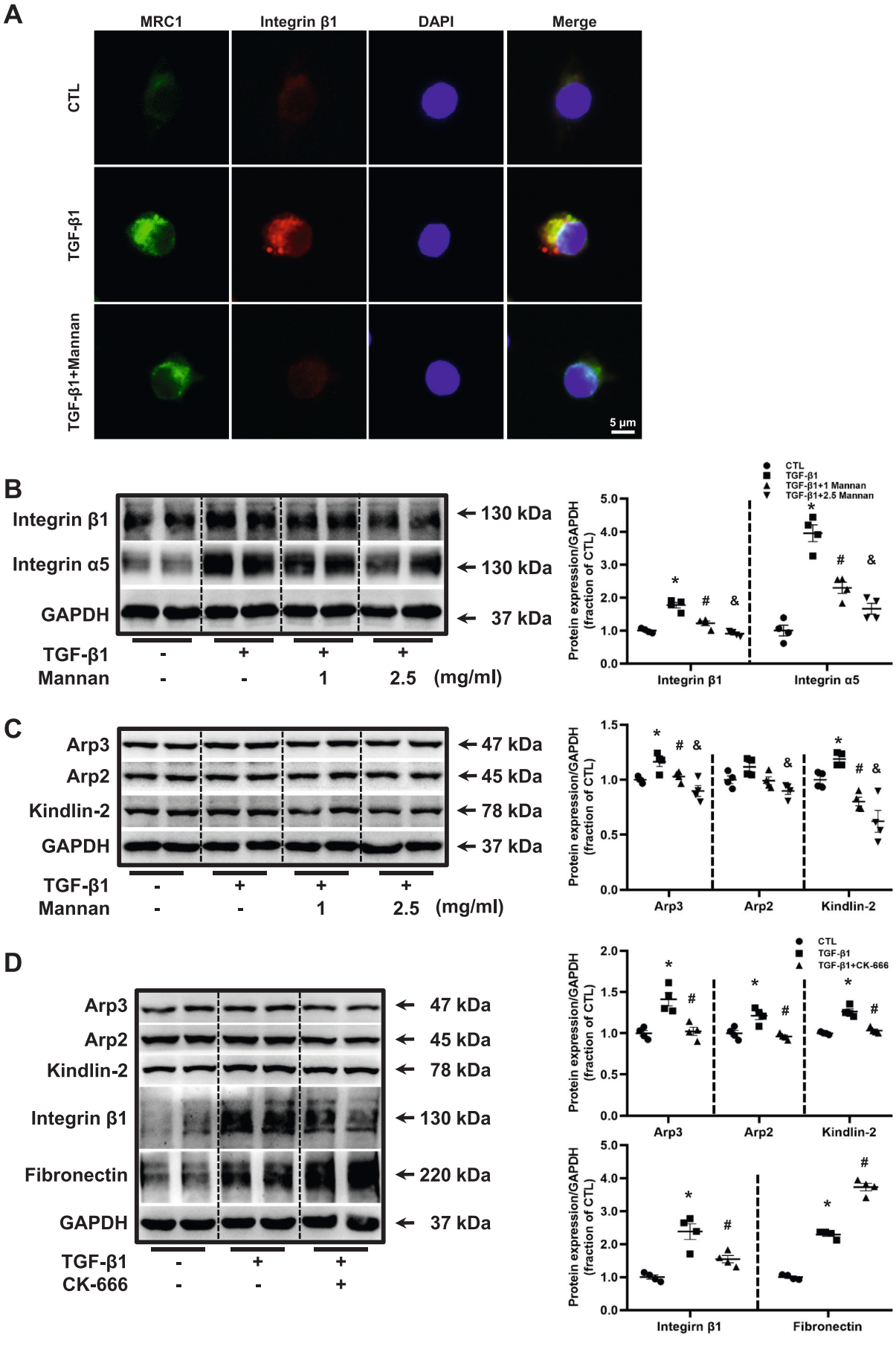


Fig. 5. MRC1 regulated integrin β1 through Arp2/3-kindlin-2 signaling pathway in 7UUO or UIRI models. (A) and (B) Representative immunoblots and corresponding densitometry analyses of Arp2, Arp3 and kindlin-2 in the kidney of WT or MRC1-KO mice with or without 7UUO (A) or UIRI (B). (C) and (D) mRNA levels of integrin β1, Arp2, Arp3 and kindlin-2 in the kidney of WT or MRC1-KO mice with or without 7UUO (C) or UIRI (D). (E) and (F) mRNA levels of AP-2, clathrin, dynamin and cortactin in the kidney of WT or MRC1-KO mice with or without 7UUO (E) or UIRI (F). Data are shown as mean ± SEM (n = 6 in each group), statistical analyses were performed using one-way ANOVA, *P < 0.05, when compared with WT-Sham group. #P < 0.05, when compared with WT-7UUO or UIRI group, respectively. WT, wild type; KO, knockout; 7UUO, unilateral ureteral obstruction for 7 days; UIRI, unilateral ischemia reperfusion injury.



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Fig. 6. MRC1 regulated integrin β1 through Arp2/3-kindlin-2 signaling pathway in TGF-β1-induced pro-fibrotic response in RAW264.7 cells. (A) Immunofluorescence of MRC1 and integrin β1 in RAW264.7 cells pretreated with mannan for 1 h followed by TGF-β1 treatment for 24 h. (B) Representative immunoblots and corresponding densitometry analyses of integrin β1 and integrin α5 protein abundance in RAW264.7 cells pretreated with mannan for 1 h followed by TGF-β1 treatment for 24 h. (C) Representative immunoblots and corresponding densitometry analyses of Arp2, Arp3 and kindlin-2 in RAW264.7 cells pretreated with mannan for 1 h followed by TGF-β1 treatment for 24 h. (D) Representative immunoblots and corresponding densitometry analyses of Arp2, Arp3, kindlin-2, integrin β1 and fibronectin in RAW264.7 cells pretreated with CK-666 for 1 h followed by TGF-β1 treatment for 24 h. Data are shown as mean ± SEM (n = 4 in each group), statistical analyses were performed using one-way ANOVA, *P < 0.05, when compared with CTL group. #P < 0.05, when compared with TGF-β1 group. CTL, control; TGF-β1, transforming growth factor-β1; CK-666, inhibitor of Arp2/3.

experimental evidence that MRC1 may prevent renal fibrogenesis.

MRC1 is mainly expressed in macrophages [37], and previous studies demonstrated that M2-like macrophages represented the principal cell type responsible for collagen turnover [38]. In RAW264.7 cells, MRC1 inhibition by pretreatment with mannan dramatically aggravated TGF-β1-induced increases in fibronectin and phosphorylation of Smad2/3

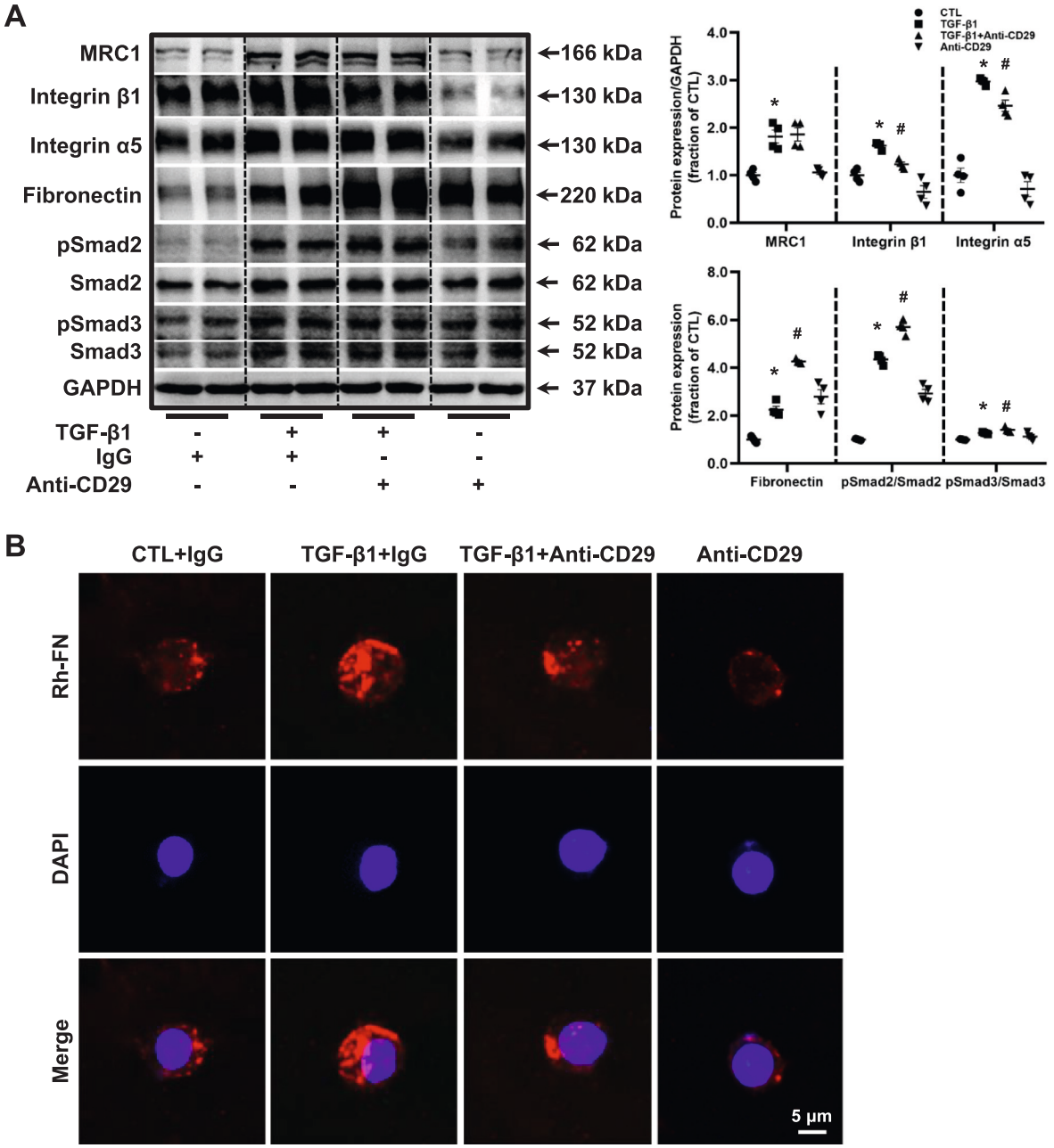


Fig. 7. Inhibition of integrin β1 reduced the endocytosis of fibronectin and aggravated TGF-β1-induced pro-fibrotic response in RAW264.7 cells. (A) Representative immunoblots and corresponding densitometry analyses of MRC1, integrin β1, integrin α5, fibronectin, pSmad2/3 and Smad2/3 protein abundance in RAW264.7 cells pretreated with Anti-CD29 for 1 h followed by TGF-β1 treatment for 24 h. (B) Representative fluorescence microscopy images of RAW264.7 cells pretreated with Anti-CD29 for 1 h followed by TGF-β1 treatment for 24 h and then incubated with Rh-FN. Data are shown as mean ± SEM (n = 4 in each group), statistical analyses were performed using one-way ANOVA, *P < 0.05, when compared with CTL group. #P < 0.05, when compared with TGF-β1 group. CTL, control; TGF-β1, transforming growth factor-β1; Anti-CD29, Integrinβ1 inhibitory antibody; Rh-FN, Rhodamine fibronectin.

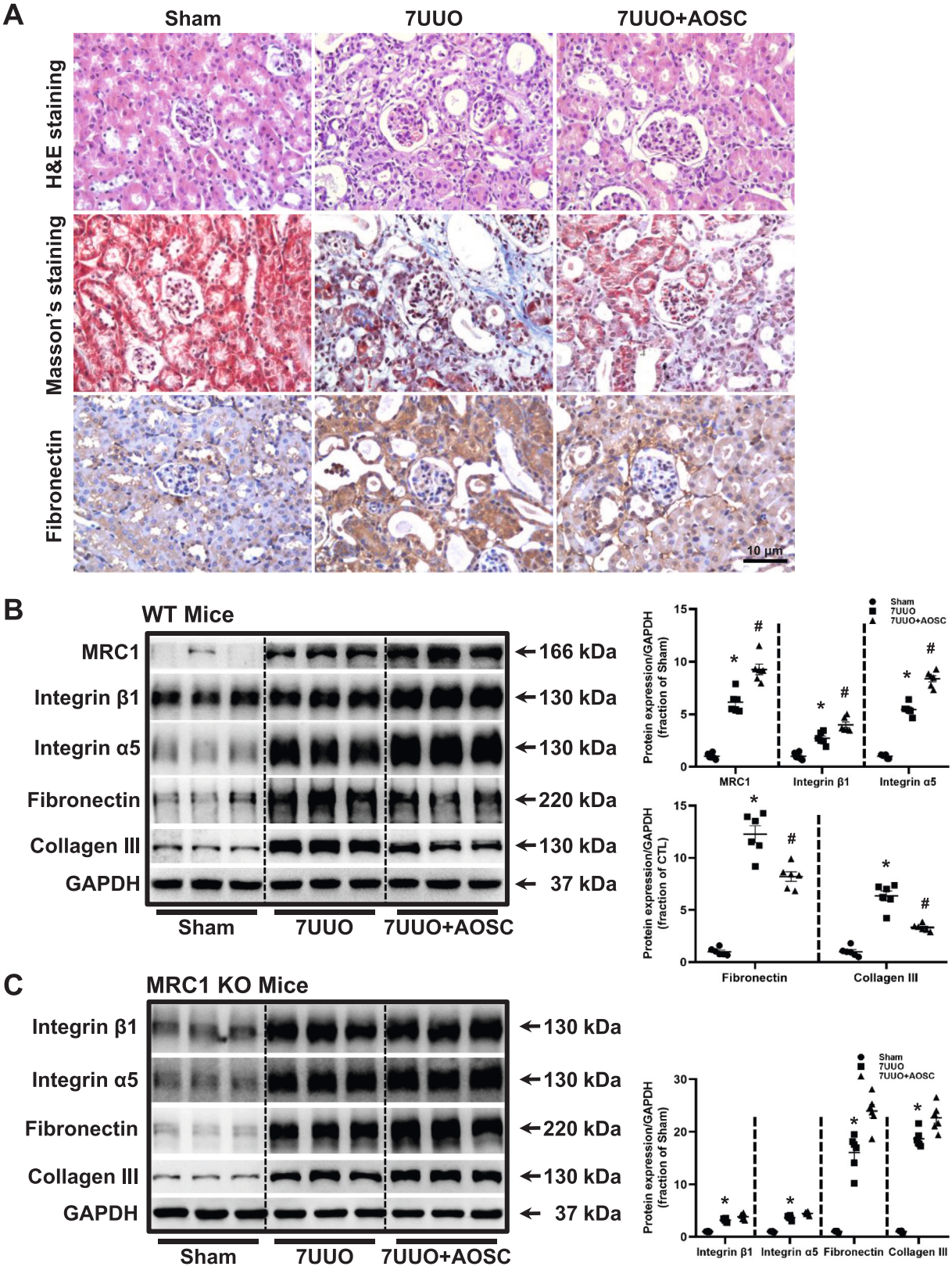


Fig. 8. AOSC up-regulated MRC1 protein expression and attenuated renal fibrosis induced by 7UUO. (A) Photomicrographs of hematoxylin-eosin staining, Masson's staining and immunohistochemistry of fibronectin in the kidney of Sham, 7UUO, 7UUO + AOSC groups in mice. (B) Representative immunoblots and corresponding densitometry analyses of MRC1, integrin β1, fibronectin and collagen III protein abundance in the kidney of Sham, 7UUO, 7UUO + AOSC groups in mice. (C) Representative immunoblots and corresponding densitometry analyses of integrin β1, integrin α5, fibronectin and collagen III protein abundance in the kidney of Sham, 7UUO, 7UUO + AOSC groups in MRC1-KO mice. Data are shown as mean ± SEM (n = 6 in each group), statistical analyses were performed using one-way ANOVA, *P < 0.05, when compared with Sham group, #P < 0.05, when compared with 7UUO group. KO, knockout; 7UUO, unilateral ureteral obstruction for 7 days; AOSC, alginate oligosaccharides.

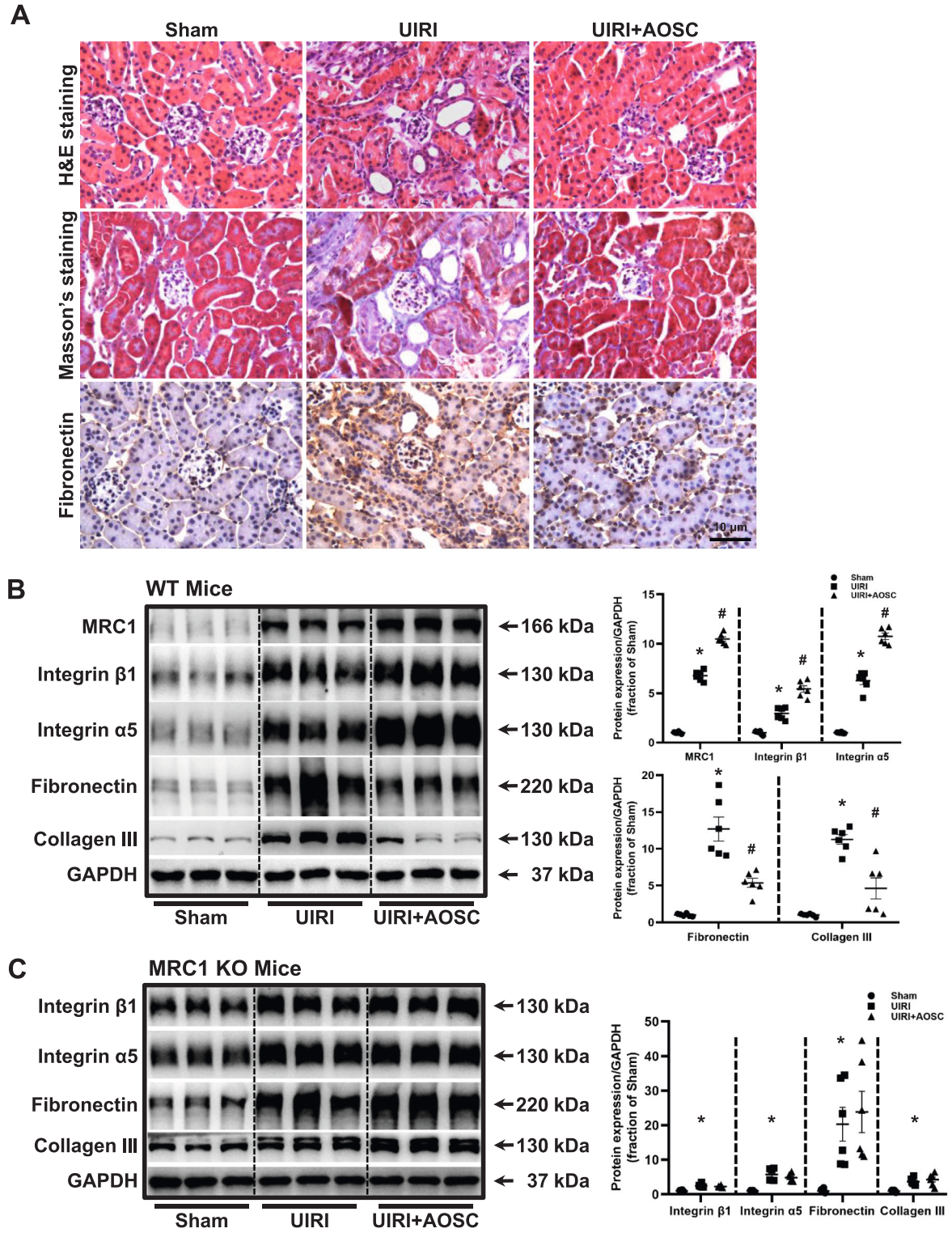


Fig. 9. AOSC up-regulated MRC1 protein expression and attenuated renal fibrosis induced by UIRI. (A) Photomicrographs of hematoxylin-eosin staining, Masson's staining and immunohistochemistry of fibronectin in the kidney of Sham, UIRI, UIRI+AOSC groups in mice. (B) Representative immunoblots and corresponding densitometry analyses of MRC1, integrin β1, fibronectin and collagen III protein abundance in the kidney of Sham, UIRI, UIRI+AOSC groups in mice. (C) Representative immunoblots and corresponding densitometry analyses of integrin β1, integrin α5, fibronectin and collagen III protein abundance in the kidney of Sham, UIRI, UIRI+AOSC groups in MRC1-KO mice. Data are shown as mean ± SEM (n = 6 in each group), statistical analyses were performed using one-way ANOVA, *P < 0.05, when compared with Sham group, $^{\#}$P < 0.05, when compared with UIRI group. KO, knockout; UIRI, unilateral ischemia reperfusion injury; AOSC, alginate oligosaccharides.

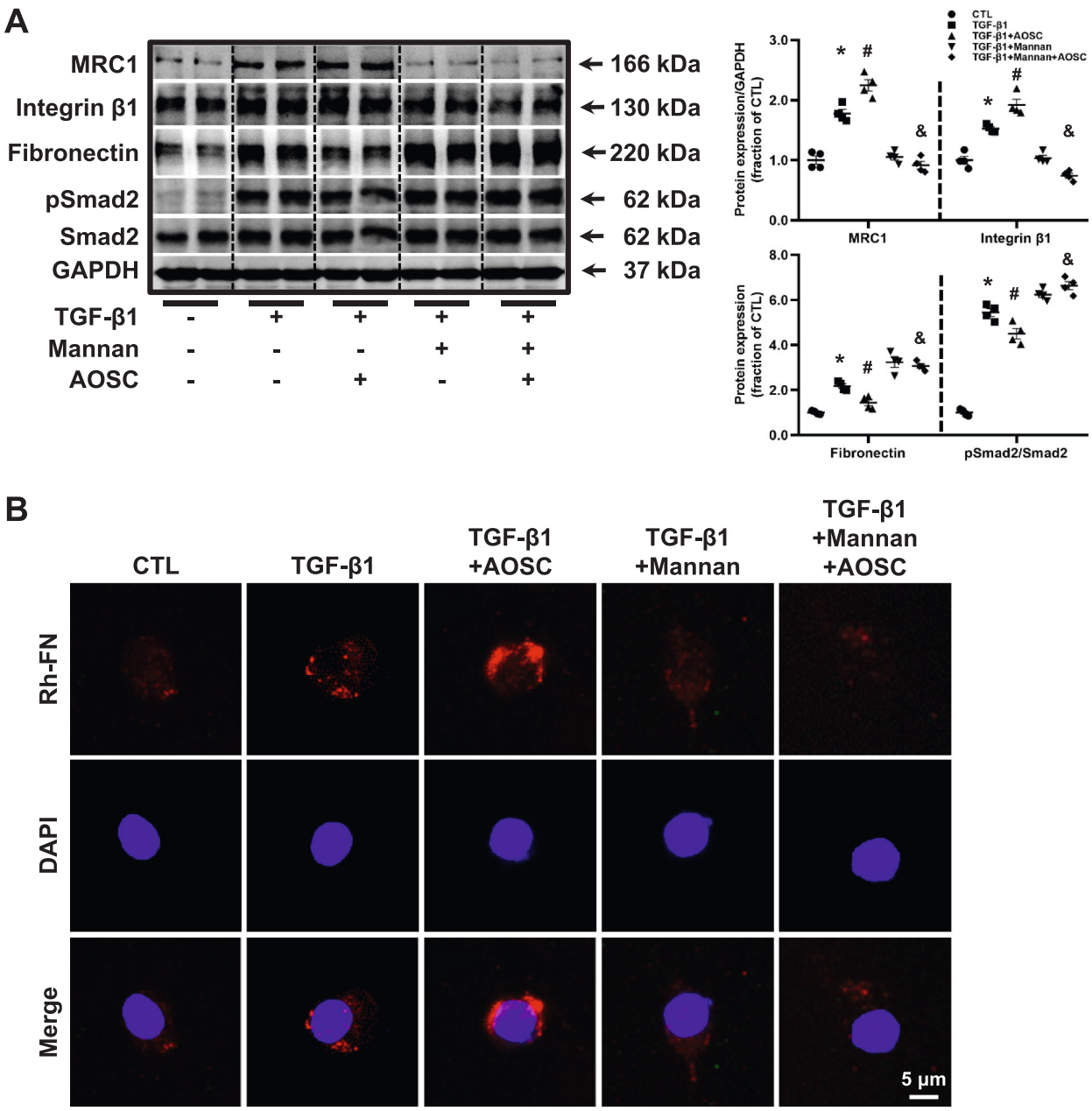


Fig. 10. AOSC up-regulated MRC1 protein expression and alleviated pro-fibrotic response in RAW264.7 cells. (A) Representative immunoblots and corresponding densitometry analyses of MRC1, integrin β1, fibronectin, pSmad2 and Smad2 protein abundance in RAW264.7 cells pretreated with AOSC for 1 h and mannan for 30 min, followed by TGF-β1 treatment for 24 h. (B) Representative fluorescence microscopy images of RAW264.7 cells pretreated with AOSC for 1 h and mannan for 30 min, followed by TGF-β1 treatment for 24 h and incubated with Rh-FN for 1 h. Data are shown as mean ± SEM (n = 4 in each group), statistical analyses were performed using one-way ANOVA, *P < 0.05, when compared with CTL group. #P < 0.05, when compared with TGF-β1 group. &P < 0.05, when compared with TGF-β1 + AOSC group. CTL, control; TGF-β1, transforming growth factor-β1; Rh-FN, Rhodamine fibronectin; AOSC, alginate oligosaccharides.

that is known a critical molecule mediating fibrotic responses. Kidney fibrogenesis is a dynamic process orchestrated by fibroblast activation, tubular atrophy, tissue inflammation and microvascular rarefaction, a complex and intimate interplay among fibroblasts, tubular epithelial cells, infiltrating inflammatory cells and vascular endothelial cells [39]. TGF-β1 markedly induced increases in protein expression of MRC1 in HK2 and NRK—49F, however, the protein abundance of fibrotic markers fibronectin remained unchanged after MRC1 inhibition by mannan. Consistent with this, the immunohistochemistry staining of F4/80 (Supplementary data Fig. S3) showed less cell number of macrophages in MRC1-KO mice with 7UUO or UIRI than WT mice. MRC1 is mainly expressed on macrophages, and involved in multiple biological pathways [14]. Although we also detected MRC1 expression on fibroblasts and tubular epithelial cells in vitro, its modulatory role in kidney fibrosis is still unclear. Our data indicated that MRC1 in fibroblasts or HK2 cells was unlikely crucial in mediating pro-fibrotic responses induced by TGF-β1. Therefore, macrophages may probably be one of the major contributors in kidney fibrogenesis in association with altered MRC1 expression.

Fibronectin is the major components of ECM and the first one to be deposited in renal interstitial fibrosis, which seems to be particularly important in the progression of fibrosis [40]. Fibronectin uptake could be determined by measuring the fluorescence intensity of rhodamine-labeled intracellular puncta as shown in previous studies [29]. Most studies to date focused on the role of MRC1 to internalize collagen [41], it looks MRC1 is also involved in uptake of other matrix proteins like

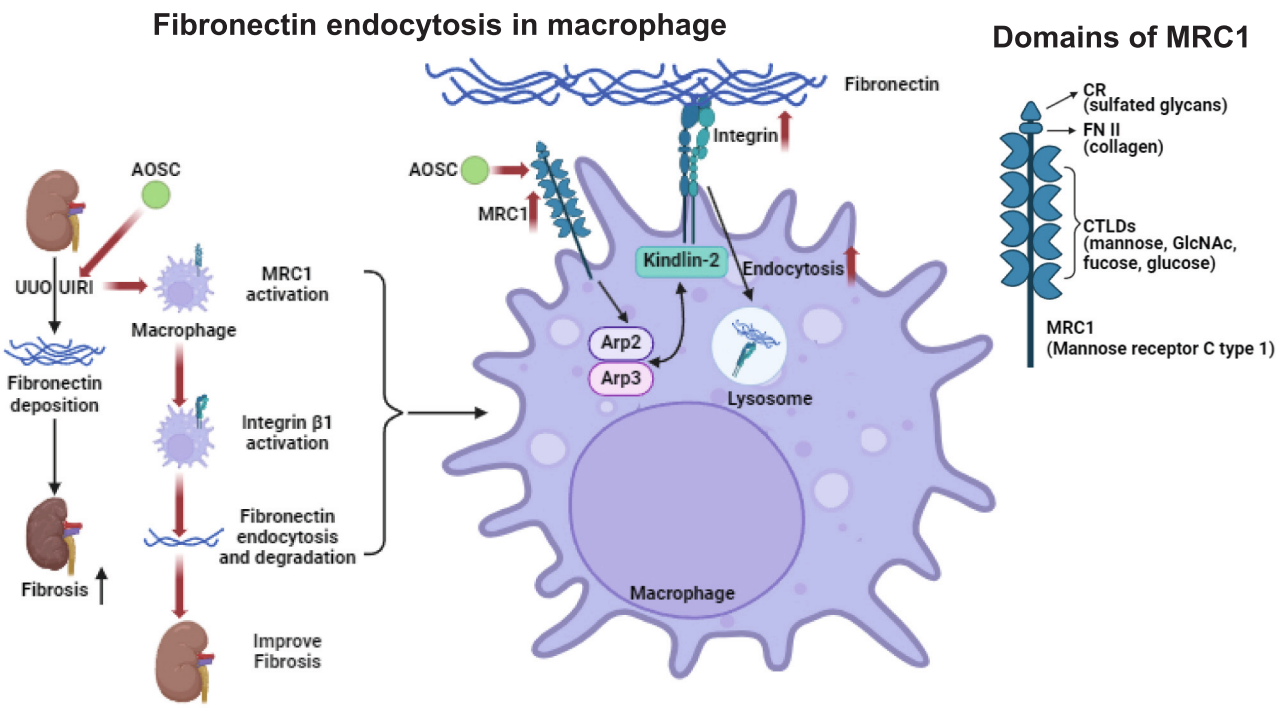


Fig. 11. The proposed mechanism of MRC1 preventing kidney fibrosis. MRC1 might promote integrin β1-mediated fibronectin endocytosis through the Arp2/3-kindlin-2 signaling pathway, and AOSC could ameliorate renal fibrosis likely through interacting with MRC1.

fibronectin. Interestingly, fibronectin endocytosis was dramatically decreased after MRC1 inhibition by mannan in RAW264.7 cells, indicating that MRC1 likely mediated fibronectin internalization. Meanwhile, fibronectin protein expression was increased consistently after MRC1 inhibition in vivo and in vitro. These data indicate that MRC1 is involved in fibrogenesis likely through modulating ECM turnover. It should be noted that MRC1 inhibition was associated with activation of TGF-β1/Smad2/3 and increased fibronectin expression, while MRC1 also mediates the endocytosis and degradation of fibronectin through integrin β1. Kidney fibrosis is a dynamic process depends on a balance between ECM deposition and degradation. Fibronectin synthesis is increased after TGF-β1 treatment, at the same time endocytosis of fibronectin initiates [3]. However, at this stage endocytosis of fibronectin could not counteract the its synthesis completely, as macrophages tend to undergo pro-fibrotic phenotype.

Evidence suggested that collagen phagocytosis via integrin was the primary cellular pathway [8]. Integrin β1 is an additional type of ECM receptors expressed in almost all cell types, which could bind to fibronectin and mediates its endocytosis [42]. Upregulation of integrin β1 induced by 7UUO or UIRI were markedly suppressed in the MRC1-KO mice. MRC1 inhibition also reduced expression of integrin β1 in TGF-β1-treated RAW264.7 cells. These findings were coincidentally associated with decreased fibronectin endocytosis. There is no identified classical intracellular signaling pathway about MRC1 [43]. It remains unknown how MRC1 interacts to integrin. A previous study reported that MRC1 could maintain the stability of Arp2/3 complex, which is believed to promote phagocytosis [32]. Arp2/3 complex interacts with kindlin-2 [33], a protein that activates integrin β1 by binding to its intracellular domain [34]. In 7UUO and UIRI models, the expression of Arp2, Arp3 and kindlin-2 protein in the kidney were decreased after MRC1 gene knockout. Consistently, the Arp2/3-kindlin-2 signaling pathway was also repressed upon MRC1 inhibition in RAW264.7 cells. To validate the role of integrin β1 in fibronectin internalization, we used anti-CD29 to serve as integrin β1 inhibitory antibody [44]. Integrin β1 inhibition by anti-CD29 clearly blocked the endocytosis of fibronectin and exacerbated the pro-fibrotic responses in RAW264.7 cells. Integrin β1 inhibition by anti-CD29 had no significant effect on the expression of MRC1, whereas MRC1 inhibition reduced the expression of integrin β1, which, again, indicated that integrin β1 was the downstream target of MRC1 signaling. It seems reasonable to assume that MRC1 regulates integrin β1 likely through Arp2/3-kindlin-2 signaling pathway, and thus modulates fibronectin endocytosis (Fig. 11).

It should be noted that renal tubular cells play important roles in initiation and development of fibrosis. Kidney fibrosis is closely linked to the severity of initial renal injury. MRC1 expressed in tubular cells may thus be involved in kidney fibrosis associated with tubular injury. As reported, the early stages of the fibrotic process are characterized by complex inflammatory events [45]. Our preliminary data showed that NF-κB-associated inflammatory signaling was activated by TGF-β1 in HK2 cells, while MRC1 inhibition failed to prevent the pro-inflammatory responses (*unpublished data*), indicating a possibility that a role of MRC1 in tubular cells is quite different from that in macrophages when stimulated by TGF-β1. However, in the other study from our group (Acta Pharm Sin, *in press*), MRC1 inhibition by mannan indeed attenuated pro-inflammatory responses in HK2 cells subjected to hypoxia/reoxygenation (H/R), which is, theoretically, supposed to prevent potential development of fibrosis. In the current study, we focused on macrophages-mediated mechanisms in renal fibrosis, the in vivo animal models were established in mice with global MRC1 gene knockout. To better understand the involvement of other types of kidney cells, conditional MRC1 knockout in macrophages or tubular cells may be needed in future studies.

Alginate oligosaccharides are the degradation products of alginate, a natural extract of brown algae [46]. In a rat model of salt-sensitive hypertension, sodium alginate oligosaccharides significantly attenuated hypertensive glomerular sclerosis [47]. AOSC can also alleviated kidney fibrosis in D-gal-induced aging mice [25]. In kidney fibrosis models induced by 7UUO and UIRI in mice, a significant remission of interstitial fibrosis was found after AOSC treatment. The results of in vitro model also indicated that AOSC inhibited TGF-β1-induced pro-fibrotic responses by promoting fibronectin endocytosis in macrophages (Fig. 10). Recent studies indicated AOSC prevented mucositis likely through

interacting with MRC1 [48]. In the present study, AOSC treatment upregulated the protein expression of MRC1 and integrin β1, which was, however, prevented by MRC1 knockout in vivo or inhibition in vitro. Taken together, these data suggested that AOSC prevented renal fibrosis through MRC1 signaling pathway, and it is probably a therapeutic strategy to delay the progress of kidney fibrosis. Our study suggests that AOSC protects against renal fibrosis via MRC1, it remains unclear whether AOSC also affects fibronectin expression in tubular cells, since renal tubular injury is a key driver of fibrosis, further investigation into how AOSC influences fibronectin regulation in tubular cell types would add valuable mechanistic insights. The current study suggested an interaction between AOSC and MRC1 in the macrophages, while the mechanism behind was not investigated. In the other study from our group, however, showed that AOSC interacted with E725, N727, E733, T743, S745 and N747 of CLTD domain of human MRC1 through hydrogen bonds, by molecular docking. The thermal stability of human MRC1 in the AOSC-treated HK2 cells was enhanced with the increased temperatures in a AOSC concentration-dependent pattern, suggesting a potentially direct interaction between AOSC and MRC1 (Acta Pharm Sin, *in press*).

In conclusion, the current study demonstrated that MRC1 might promote integrin β1-mediated fibronectin endocytosis through the Arp2/3-kindlin-2 signaling pathway, and AOSC could ameliorate renal fibrosis likely through interacting to MRC1. Our study suggests that MRC1 may be an important target for renal fibrosis, and AOSC could be a beneficial therapeutic drug for preventing or delaying the development of renal fibrosis.

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CRediT authorship contribution statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data presented in this study are available on request from the corresponding author.

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